CE205 Databases and Information Retrieval

Class 1

Objective: The purpose of these problems is to make sure that you can identify the update anomalies and determine functional dependencies in a given relation.

Problem 1: The relation **AIRPORT** captures the data about daily departing flights

AIRPORT

Date	AirlineID	AirlineName	TerminalID	NumberOfGates	NumberOfDepartingFlights
11-Dec	UA	United	A	20	34
11-Dec	NW	Northwest	A	20	17
11-Dec	AA	American	A	20	11
11-Dec	DL	Delta	B	15	20
11-Dec	JB	Jet Blue	B	15	6
12-Dec	UA	United	A	20	29
12-Dec	DL	Delta	B	15	20
12-Dec	SWA	Southwest	C	15	17

Consider the following Assumptions:

(1) Each Airline has Unique Airline ID, Airline Name.
(2) Each terminal has unique Terminal ID, fixed NumberOfGates.
(3) Each airline has one permanently assigned terminal.
(4) Each Terminal can have multiple airlines.
(5) Each day flights depart from the airport for every airline.

1(a) Give an example for each update anomaly in the relation AIRPORT.

Answer:

Insertion anomaly:
You cannot add a new terminal 'D' and NumberOfGates '10' without adding an airline associated with it.

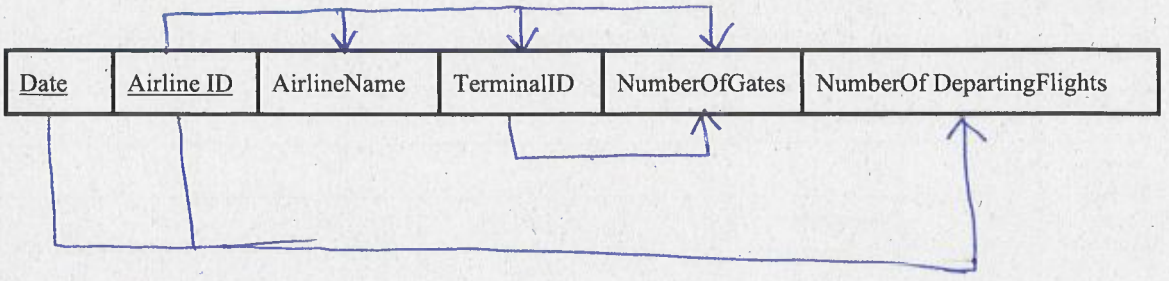
Deletion anomaly:
If you were to delete airline 'SWA' the information about terminal 'C' would also be deleted.

Modification Anomaly:
Changing the NumberOfGates value for terminal A from 20 to 25 would require making a change in four different records.

1(b) Find functional dependencies in the relation Airport using the given assumptions and represent them graphically. Also, write FDs in the standard form.

Answer:

Graphical Representation of FDs:



Standard Form of FDs:

Airline ID → AirlineName

Airline ID → TerminalID

Airline ID → NumberOfGates

TerminalID → NumberOfGates

Date, Airline ID → NumberOf DepartingFlights

Problem 2: Consider the following relation SHIPMENTS with sample data

SHIPMENTS

A	B	C	D	E	F	G
ShipmentID	**ShipmentDate**	**TruckID**	**TruckType**	**ProductID**	**ProductType**	**Quantity**
111	1-Jan	T-111	Semi-Trailer	A	Tire	180
111	1-Jan	T-111	Semi-Trailer	B	Battery	120
222	2-Jan	T-222	Van Truck	C	SparkPlug	10,000
333	3-Jan	T-333	Semi-Trailer	D	Wipers	5,000
333	3-Jan	T-333	Semi-Trailer	A	Tire	200
444	3-Jan	T-222	Van Truck	C	SparkPlug	25,000
555	3-Jan	T-111	Semi-Trailer	B	Battery	180

Consider the following Assumptions:

- *The SHIPMENTS Table for a car part company captures the data about car part shipments.*
- *Each shipment has a unique Shipment ID and a Shipment Date.*
- *Each shipment is shipped on one truck.*
- *Each truck has a unique Truck ID and a Truck Type.*
- *Each shipment can ship multiple products.*
- *Each product has a unique Product ID and a Product Type.*
- *Each quantity of a product shipped in a shipment is recorded in a table in the column Quantity.*

2(a) Using the SHIPMENTS table, describe an example that illustrates the insertion anomaly.

2(b) Using the SHIPMENTS table, describe an example that illustrates the deletion anomaly.

2(c) Using the SHIPMENTS table, describe an example that illustrates the modification anomaly.

2(d) Find functional dependencies in the relation SHIPMENTS using the given assumptions and represent them graphically. Also, write FDs in the standard form.

Answers:

2(a) Insertion anomaly:
You cannot add a new truck 'T-444' without adding a new shipment.

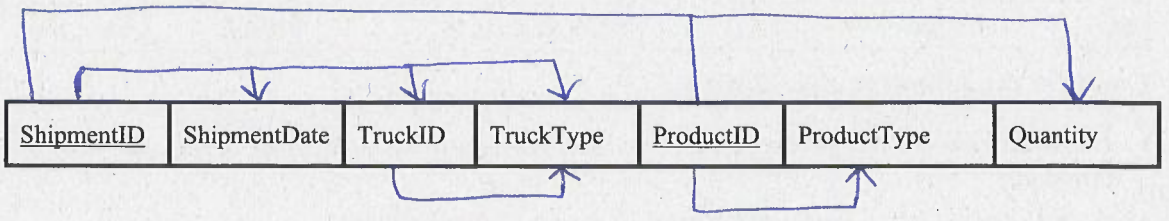
2(b) Deletion anomaly:
If you were to delete shipment '333' of 3-Jan the information about product 'D' would also be deleted.

2(c) Modification Anomaly:

Changing the TruckType 'Semi-Trailer' would require making a change in five different records.

2(d)

Graphical Representation of FDs:



Standard Form of FDs:

ShipmentID → ShipmentDate

ShipmentID → TruckID

ShipmentID → TruckType

TruckID → TruckType

ProductID → ProductType

Product ID

ShipmentID, ~~ShipmentDate~~ → Quantity